



IILM UNIVERSITY

GREATER NOIDA, UTTAR PRADESH

DATA SCIENCE INTERNSHIP

Iris Flower Classification Using Machine Learning



PROGRAM & SEMESTER

B.Tech CSE — 7th Semester



ORGANIZATION

OASIS Infobyte

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Objectives & Scope

Flower-species classification is a classic entry point into supervised ML: simple enough to reason about by hand, yet rich enough to demonstrate a complete modelling pipeline on Fisher's 1936 Iris dataset.



Inspect & Validate

Confirm dataset integrity — shape, dtypes, missing values, class balance



Explore Visually

Understand feature distributions and class separability through EDA



Reason About Features

Judge feature usefulness qualitatively, then confirm quantitatively



Prepare Rigorously

Stratified split + scaling that avoids data leakage



Train Diverse Models

Linear, distance-based, tree-based and ensemble algorithms



Select & Justify

Compare on more than accuracy; recommend a best model with evidence

Dataset Overview

150

Total Samples

4

Numeric Features

3

Balanced Species

0

Missing Values

Mean Feature Values by Species

Species	Sepal L	Sepal W	Petal L	Petal W
Setosa	5.01	3.43	1.46	0.25
Versicolor	5.94	2.77	4.26	1.33
Virginica	6.59	2.97	5.55	2.03

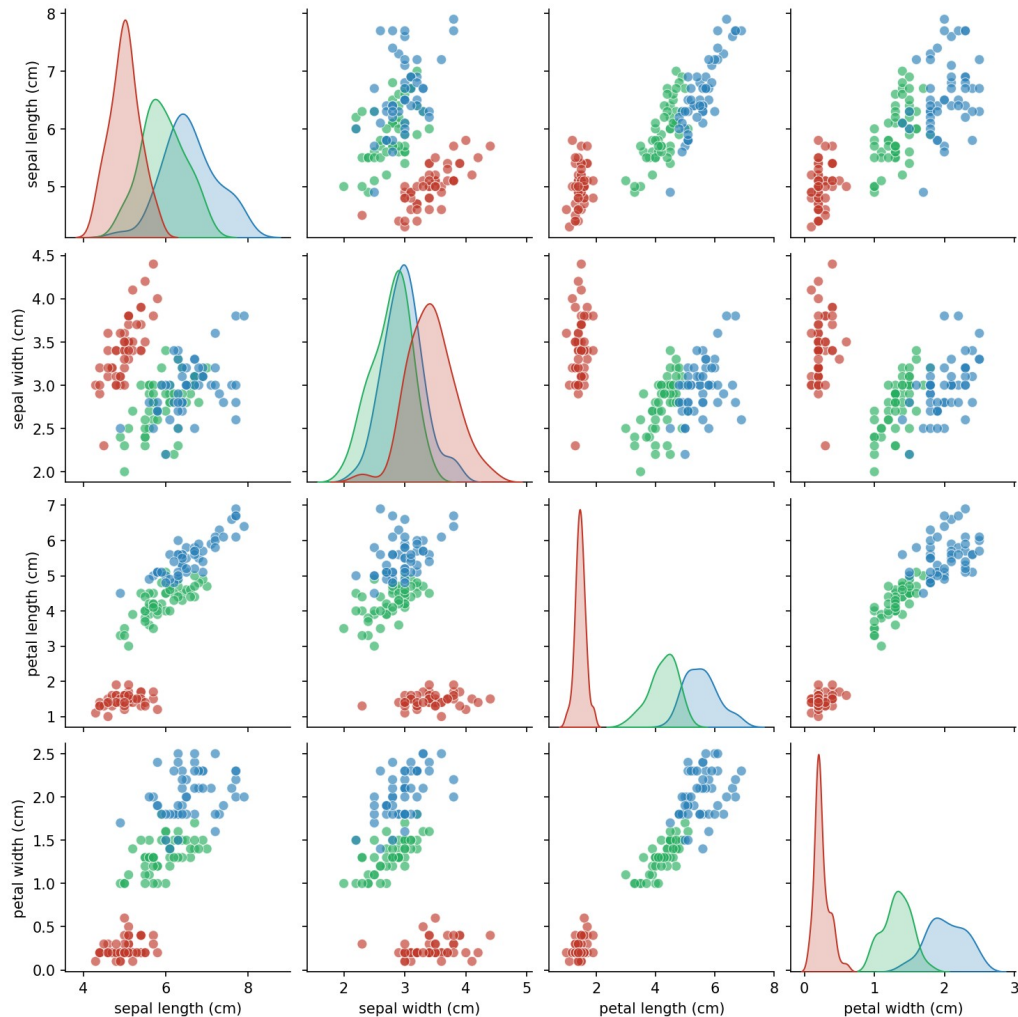
Descriptive Statistics (cm)

Feature	Mean	Std	Min	Max
Sepal length	5.84	0.83	4.30	7.90
Sepal width	3.06	0.44	2.00	4.40
Petal length	3.76	1.77	1.00	6.90
Petal width	1.20	0.76	0.10	2.50

Perfectly balanced across species — accuracy is a fair, unbiased headline metric with no majority class to exploit and no need for resampling.

Setosa is smallest in petal size but largest in sepal width — a hint that sepal width alone is a weak, counter-intuitive predictor.

Pair Plot: Feature Relationships



Setosa is isolated

Forms a tight, completely separate cluster in almost every feature pair.



Petal features separate best

Versicolor and Virginica overlap only modestly, mostly in sepal-based views.



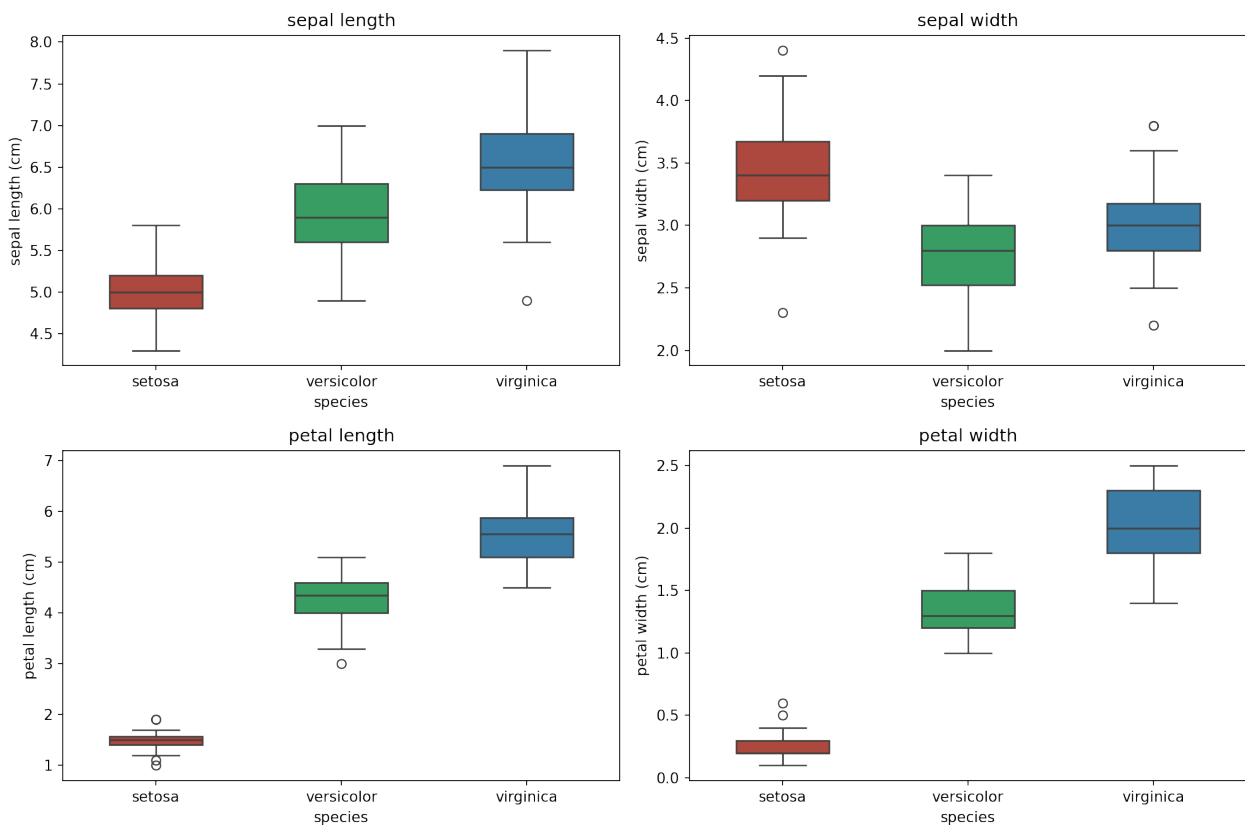
Sepal pairs overlap more

Weaker signal — consistent with lower discriminative ability.

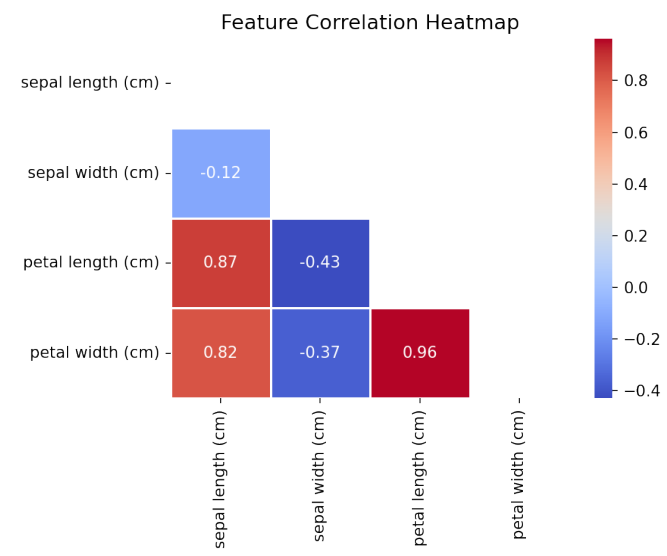
Species legend

- Setosa
- Versicolor
- Virginica

Box Plots & Feature Correlation



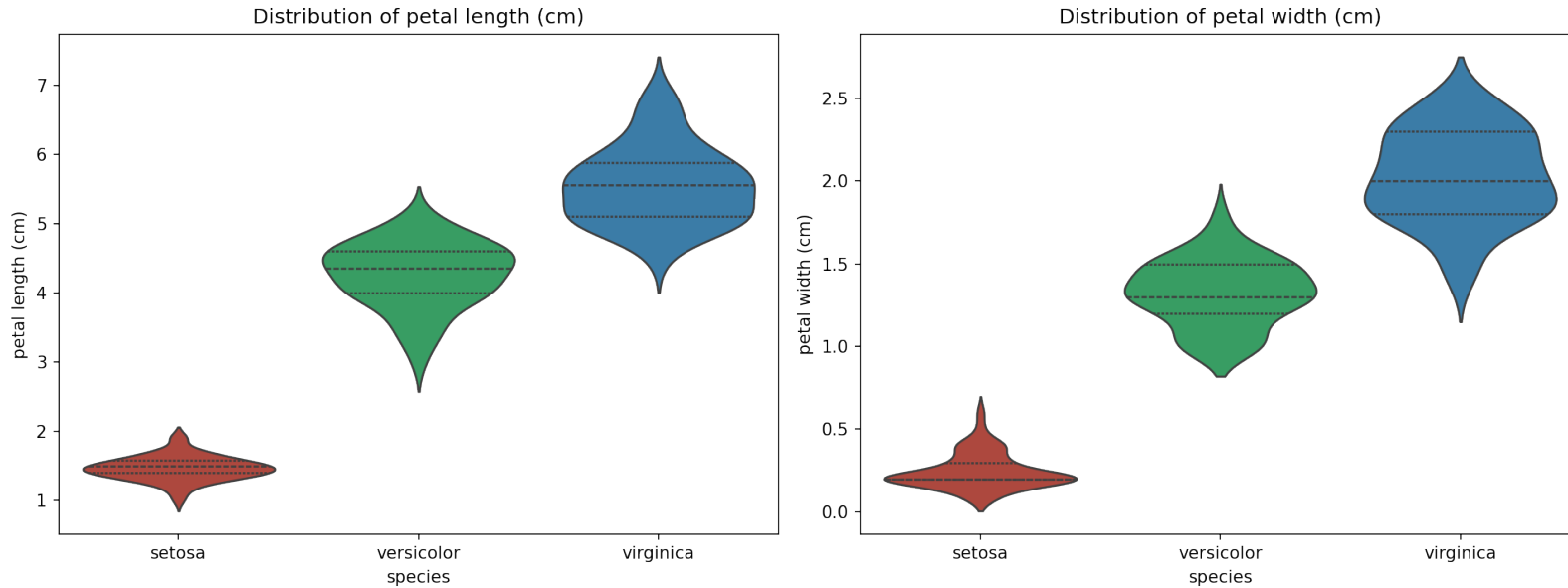
Per-feature spread by species



Correlation between numeric features

Petal length/width are strongly correlated with each other and with sepal length. Sepal width stands apart — weak or slightly negative correlation with the rest.

Violin Plots & EDA Summary

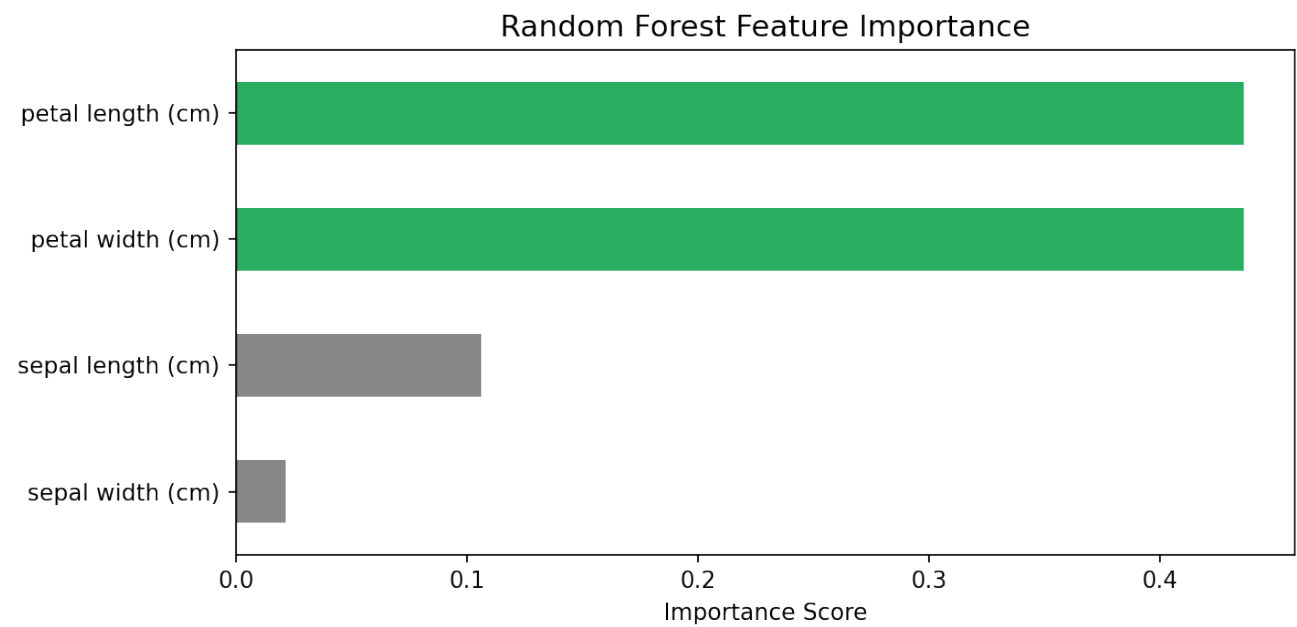


Petal length & petal width: three distinct, mostly non-overlapping density regions

EDA TAKEAWAYS

- Setosa is linearly separable on almost every feature
- Versicolor / Virginica overlap more, but remain largely distinguishable on petal dimensions
- Petal length & width are the most discriminative, correlated pair
- Sepal width is the weakest, most independent feature

Random Forest Feature Importance



Preliminary full-dataset fit, used for analysis only — the held-out train/test split in Section 6 is separate and does not leak information.

Qualitative Feature Assessment

Feature	Ability	Reasoning
Petal length	High	Nearly zero overlap between species
Petal width	High	Tight, well-separated clusters
Sepal length	Moderate	Setosa separates; others overlap
Sepal width	Low	Heavy overlap across all species

Petal length and width together account for ~87% of the model's decision signal. No features were removed — even weak ones can add marginal benefit on a set this small.

Data Preparation Pipeline



STEP 1

Split

80 / 20 stratified train-test split

120 train • 30 test samples, 10 per class in the test set



STEP 2

Standardise

$$z = (x - \mu) / \sigma$$

Scaler fit on training data only, then applied to test data



STEP 3

Avoid Leakage

No test-set information used to fit the scaler

Critical for KNN (distance-based) & Logistic Regression



Why it matters: without stratification, a single unlucky split could under-represent a class in the test set, making the reported accuracy unreliable on a dataset this small.

Four Classification Algorithms



Logistic Regression

Linear • Baseline

Softmax over a weighted feature sum.
max_iter=200. Interpretable coefficients.



K-Nearest Neighbours

Distance-based

Majority vote of k=5 nearest neighbours (Euclidean). No boundary-shape assumptions.



Decision Tree

Rule-based

Greedy recursive splits. max_depth=4.
Reads directly as if/else rules.



Random Forest

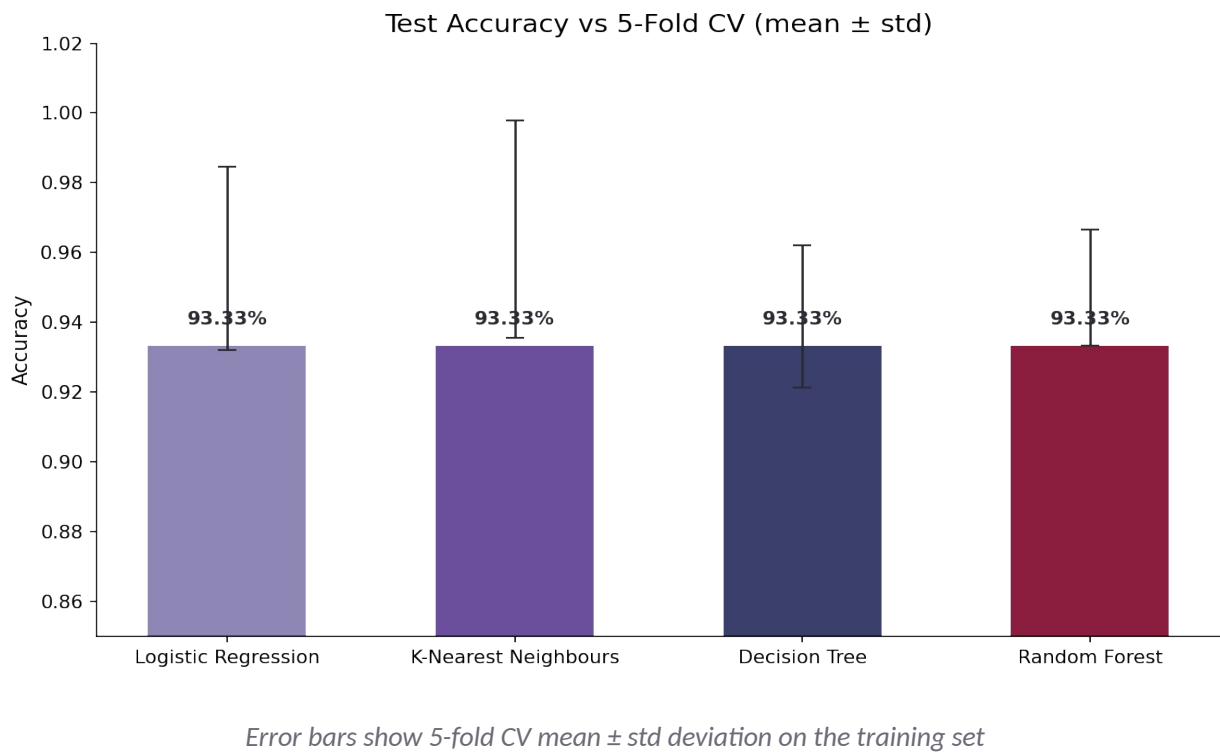
Ensemble

100 trees, max_depth=5, bootstrapped
+ random feature subsets, majority vote.

All four are trained on the identical scaled training data, so results are directly comparable across four different modelling philosophies.

RESULTS

Accuracy & Cross-Validation



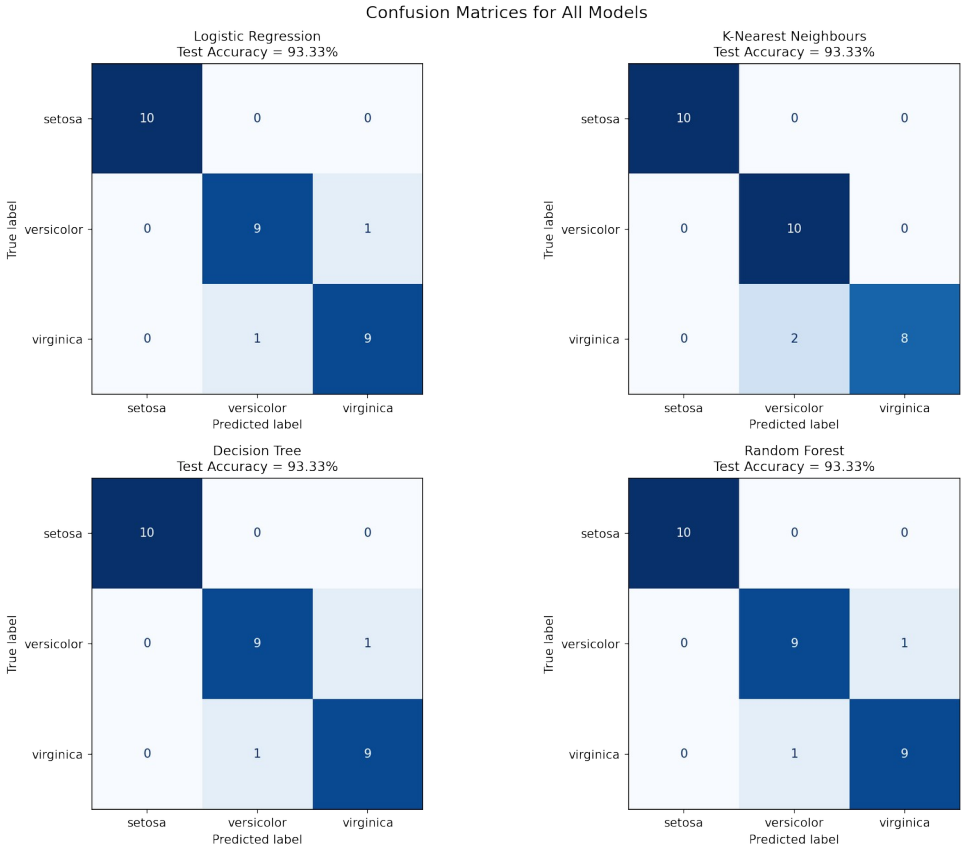
Model	Test Acc.	CV Mean	CV Std
Logistic Regression	93.33%	95.83%	0.026
K-Nearest Neighbours	93.33%	96.67%	0.031
Decision Tree	93.33%	94.17%	0.020
Random Forest	93.33%	95.00%	0.017

All four reach an identical 93.33% test accuracy (28/30) — the test set alone can't separate them, so 5-fold cross-validation breaks the tie.

KNN has the highest CV mean (96.67%) but also the most variability ($\sigma=0.031$).

Random Forest has the lowest CV std (0.017) — the most consistent across folds.

Confusion Matrix Analysis



Setosa: zero errors

All 10 Setosa test samples are classified correctly by every single model.



Only Versicolor ↔ Virginica confusion

Every misclassification, on every model, occurs only between these two species.



KNN errs differently

All 10 Versicolor correct, but 2 Virginica pulled toward Versicolor — a distinct, local-neighbourhood error pattern.

Relying on one aggregate accuracy figure would hide this — the confusion matrix reveals a real behavioural difference between models with identical test accuracy.

What the Results Really Show

The Iris dataset is close to linearly separable, so even a simple linear model performs on par with more flexible non-linear ones — the ceiling on accuracy is set by genuine overlap between two species, not by model choice.



Two independent angles agree

Feature importance (Sec. 5) and confusion matrices (Sec. 7) both point to the same story: petal measurements dominate, and errors concentrate only between Versicolor and Virginica.



Models differ in behaviour, not accuracy

The real difference between models is how — and how consistently — they resolve the ambiguous Versicolor/Virginica boundary, not their headline test score.



Local vs. global decision-making

KNN's neighbourhood-based predictions produce a qualitatively different error pattern than the other three models, despite an identical test accuracy.

CONCLUSION

Best Model: Random Forest



93.33% test accuracy · **0.017** lowest CV std of all models



Joint-highest test accuracy

Matches the best score of any model — 28 of 30 held-out samples correct.



Most stable across folds

Lowest cross-validation standard deviation (0.017) — generalises reliably rather than fitting one split.



Robust to weak features

As an ensemble on random feature subsets, it handles sepal width's low signal without manual elimination.



Transparent at the ensemble level

Feature importance scores explain its decisions — petal dimensions dominate — rather than acting as a black box.

Logistic Regression remains a strong, lightweight alternative — equal accuracy with simpler, directly interpretable coefficients.

Limitations & Future Work

LIMITATIONS

- Small dataset — each of 30 test samples is ~3.3% of accuracy
- Headline accuracy comes from a single 80/20 split
- No hyperparameter search — values set by reasonable defaults
- Limited to the four classic measurements only
- Results may not transfer to noisier, real-world data

FUTURE WORK



Systematic hyperparameter search (GridSearchCV) for every model



Test additional algorithms — SVM, Gradient Boosting



Visualise 2-D decision boundaries on the top two features



Repeat across multiple random seeds for a robust estimate

Tools, Tech Stack & Deliverables

TECH STACK

- **Python 3.10+**
Language
- **pandas • numpy**
Data handling
- **scikit-learn**
Machine learning
- **matplotlib • seaborn**
Visualisation
- **Jupyter Notebook**
Development

PROJECT DELIVERABLES

File	Description
iris_classification.ipynb	Full notebook — code, analysis, narrative
requirements.txt	Python package dependencies
README.md	Overview, setup instructions, checklist
*.png (7 figures)	Pairplot, boxplots, heatmap, violins, importance, confusion matrices, accuracy chart

All figures and numeric results are reproducible from the notebook's exact logic, with a fixed random seed (random_state = 42).

Thank You